

```
sudiksha@localhost:~  
[sudiksha@localhost ~]$ date  
Tuesday 27 January 2026 06:03:56 PM IST  
[sudiksha@localhost ~]$ echo "God is Great"  
God is Great  
[sudiksha@localhost ~]$ cal June 2007  
      June 2007  
Su Mo Tu We Th Fr Sa  
          1  2  
 3  4  5  6  7  8  9  
10 11 12 13 14 15 16  
17 18 19 20 21 22 23  
24 25 26 27 28 29 30  
[sudiksha@localhost ~]$ who  
sudiksha seat0      2026-01-27 18:03 (login screen)  
sudiksha tty2       2026-01-27 18:03 (tty2)  
[sudiksha@localhost ~]$ who am i  
[sudiksha@localhost ~]$ whoami  
sudiksha  
[sudiksha@localhost ~]$ id  
uid=1000(sudiksha) gid=1000(sudiksha) groups=1000(sudiksha) context=unconfined_u  
:unconfined_r:unconfined_t:s0-s0:c0.c1023  
[sudiksha@localhost ~]$ tty  
/dev/pts/0
```

```
sudiksha@localhost:~  
[sudiksha@localhost ~]$
```

```
[sudiksha@localhost ~]$ ls  
abc Desktop e h linux n      Public sudi systeml x  
b Documents f j m o      raja sudi.c Templates y  
c Downloads g l Music Pictures sample.c sudiksha Videos z
```

```
sudiksha@localhost:~$ man pwd  
NAME  
pwd - print name of current/working directory  
SYNOPSIS  
pwd [OPTION]...  
DESCRIPTION  
Print the full filename of the current working directory.  
-L, --logical  
use PWD from environment, even if it contains symlinks  
-P, --physical  
avoid all symlinks  
--help display this help and exit  
--version  
output version information and exit  
If no option is specified, -P is assumed.  
Manual page pwd(1) line 1 (press h for help or q to quit)
```

```
sudiksha@localhost:~/sudiksha  
[sudiksha@localhost ~]$ man pwd  
[sudiksha@localhost ~]$ man ls  
[sudiksha@localhost ~]$ ps  
PID TTY TIME CMD  
2552 pts/0 00:00:00 bash  
2839 pts/0 00:00:00 ps  
[sudiksha@localhost ~]$ uname -a  
Linux localhost.localdomain 5.14.0-654.el9.x86_64 #1 SMP PREEMPT_DYNAMIC Fri Dec 19 08:34:59 UTC 2025 x86_64 x86_64 GNU/Linux  
[sudiksha@localhost ~]$ pwd  
/home/sudiksha  
[sudiksha@localhost ~]$  
[sudiksha@localhost ~]$ mkdir recaids  
[sudiksha@localhost ~]$ rmdir recaids  
[sudiksha@localhost ~]$ ls  
abc Desktop e h linux n      Public sudi systeml x  
b Documents f j m o      raja sudi.c Templates y  
c Downloads g l Music Pictures sample.c sudiksha Videos z  
[sudiksha@localhost ~]$ cd system1  
bash: cd: system1: Not a directory  
[sudiksha@localhost ~]$ cd sudiksha  
[sudiksha@localhost ~]$ cd sudiksha$
```

```
sudiksha@localhost:~  
2552 pts/0 00:00:00 bash  
2839 pts/0 00:00:00 ps  
[sudiksha@localhost ~]$ uname -a  
Linux localhost.localdomain 5.14.0-654.el9.x86_64 #1 SMP PREEMPT_DYNAMIC Fri Dec 19 08:34:59 UTC 2025 x86_64 x86_64 GNU/Linux  
[sudiksha@localhost ~]$ pwd  
/home/sudiksha  
[sudiksha@localhost ~]$  
[sudiksha@localhost ~]$ mkdir recaids  
[sudiksha@localhost ~]$ rmdir recaids  
[sudiksha@localhost ~]$ ls  
abc Desktop e h linux n      Public sudi systeml x  
b Documents f j m o      raja sudi.c Templates y  
c Downloads g l Music Pictures sample.c sudiksha Videos z  
[sudiksha@localhost ~]$ cd system1  
bash: cd: system1: Not a directory  
[sudiksha@localhost ~]$ cd sudiksha  
[sudiksha@localhost ~]$ cd ..  
[sudiksha@localhost ~]$ cat sample.c  
ls -> used to list all the available folder  
ls -l -> used to list all the hidden folder  
[sudiksha@localhost ~]$ cp system1 system2  
[sudiksha@localhost ~]$ rm system1  
[sudiksha@localhost ~]$
```

```
sudiksha@localhost:~/Documents
[sudiksha@localhost ~]$ cat>rec
This is Linux
cat command example
[sudiksha@localhost ~]$ cat rec
This is Linux
cat command example
[sudiksha@localhost ~]$ cp rec rec_copy
[sudiksha@localhost ~]$ cp sample.c Documents/
[sudiksha@localhost ~]$ ls
.  ..  Desktop  Downloads  Linux  Music  Pictures  Public  Templates  Videos  rec  sudi.c  sample.c  system2
[sudiksha@localhost ~]$ rm rec_copy
[sudiksha@localhost ~]$ mv sample.c sample_old.c
[sudiksha@localhost ~]$ file system2
system2: ASCII text
[sudiksha@localhost ~]$ wc rec
 2  6 34 rec
[sudiksha@localhost ~]$ who|wc -l
 2
[sudiksha@localhost ~]$ who|rec|wc -l
bash: rec: command not found...
0
```

```
sudiksha@localhost:~/Documents
[sudiksha@localhost ~]$ who|tee sample|wc -l
 2
[sudiksha@localhost ~]$ ls sample
sample
[sudiksha@localhost ~]$ ls system2
system2
[sudiksha@localhost ~]$ ls -l
total 28
drwxr-xr-x. 2 sudiksha sudiksha  6 Jan  9 09:32 .aka
drwxr-xr-x. 2 sudiksha sudiksha  6 Jan  9 09:38 .b
drwxr-xr-x. 2 sudiksha sudiksha  6 Jan  9 09:35 .c
drwxr-xr-x. 2 sudiksha sudiksha  6 Jan  5 28:58 Desktop
drwxr-xr-x. 2 sudiksha sudiksha 22 Jan 27 28:54 Documents
drwxr-xr-x. 2 sudiksha sudiksha  6 Jan  5 28:58 Downloads
drwxr-xr-x. 2 sudiksha sudiksha 39 Jan  9 09:41 .e
drwxr-xr-x. 2 sudiksha sudiksha  6 Jan  9 09:38 .f
drwxr-xr-x. 2 sudiksha sudiksha  6 Jan  9 09:38 .g
drwxr-xr-x. 2 sudiksha sudiksha  6 Jan 12 14:51 .h
drwxr-xr-x. 2 sudiksha sudiksha 30 Jan 21 16:12 .j
drwxr-xr-x. 2 sudiksha sudiksha  6 Jan 21 15:57 .l
drwxr-xr-x. 2 sudiksha sudiksha 39 Jan  7 16:39 Linux
drwxr-xr-x. 2 sudiksha sudiksha  6 Jan 21 15:57 .m
drwxr-xr-x. 2 sudiksha sudiksha  6 Jan  5 28:58 Music
```

```
sudiksha@localhost:~/Documents
[sudiksha@localhost Documents]$ chmod u+x sample.c
[sudiksha@localhost Documents]$ chmod u+x sample.c
[sudiksha@localhost Documents]$ ls -l sample.c
-rwxr-xr-x. 1 sudiksha sudiksha 88 Jan 27 20:54 sample.c
[sudiksha@localhost Documents]$ chmod 761 sample.c
[sudiksha@localhost Documents]$ ls -l sample.c
-rwxr-xr-x. 1 sudiksha sudiksha 88 Jan 27 20:54 sample.c
[sudiksha@localhost Documents]$ who|date
sudiksha seat0 2026-01-27 20:50 (login screen)
sudiksha tty2 2026-01-27 20:50 (tty2)
Tuesday 27 January 2026 09:09:33 PM IST
[sudiksha@localhost Documents]$ who|date
sudiksha seat0 2026-01-27 20:50 (login screen)
sudiksha tty2 2026-01-27 20:50 (tty2)
Tuesday 27 January 2026 09:09:48 PM IST
[sudiksha@localhost Documents]$ who|date
sudiksha seat0 2026-01-27 20:50 (login screen)
sudiksha tty2 2026-01-27 20:50 (tty2)
[sudiksha@localhost Documents]$ cat>college
AI
DS
ML
ECE
EEE
EEE
```

```
sudiksha@localhost:~/Documents
AI
DS
ML
ECE
EEE
EEE
CSE
CIVIL
MECH
ROBOTICS
BIOTECH
BIOMEDICAL
IT
IOT
[sudiksha@localhost Documents]$ head college
AI
DS
ML
ECE
EEE
EEE
CSE
CIVIL
MECH
ROBOTICS
BIOTECH
BIOMEDICAL
IT
IOT
[sudiksha@localhost Documents]$ tail college
fcp
```

```
sudiksha@localhost:~/Documents
[sudiksha@localhost Documents]$ tail college
ECE
EEE
EEE
CSE
CIVIL
MECH
ROBOTICS
BIOTECH
BIOMEDICAL
IT
IOT
[sudiksha@localhost Documents]$ ls -l|more
total 8
-rw-r--r--. 1 sudiksha sudiksha 87 Jan 27 21:13 college
-rwxr-xr-x. 1 sudiksha sudiksha 88 Jan 27 20:54 sample.c
[sudiksha@localhost Documents]$ sort college
AI
BIOMEDICAL
BIOTECH
CIVIL
CSE
DS
ECE
EEE
IOT
IT
MECH
ML
ROBOTICS
```

```
sudiksha@localhost:~/Documents
[sudiksha@localhost Documents]$ sort college
AI
BIOMEDICAL
BIOTECH
CIVIL
CSE
DS
ECE
EEE
IOT
IT
MECH
ML
ROBOTICS
[sudiksha@localhost Documents]$ nl college
 1 AI
 2 DS
 3 ML
 4 ECE
 5 EEE
 6 CSE
 7 CIVIL
 8 MECH
```

```
[sudiksha@localhost Documents]$ cut -c1 college
A
D
M
E
C
M
R
B
I
I
[sudiksha@localhost Documents]$
```

```
sudiksha@localhost:~/Documents — top
top - 21:19:02 up 28 min, 2 users, load average: 0.42, 0.10, 0.11
Tasks: 183 total, 1 running, 182 sleeping, 0 stopped, 0 zombie
Cpu(s): 1.3 us, 1.4 sy, 0.2 id, 88.3 lo, 0.9 wa, 7.4 hi, 1.8 si, 0.0 st
Mem: 1787.0 total, 386.1 free, 844.3 used, 646.2 buff/cache
Mem%: 264.4 total, 264.4 free, 844.3 used, 839.3 avail mem

PID USER      PR  NI    TID    CPU   SPMs  MEM%  SPMs  COMMAND
1909 sudiksha 20  0 5409328 89788 127540 5 5.2 17.0 0.13.51 gnome-shell
2199 root      0  0 0 0 0 0 0 0 0.04.41 kworker/0:1-events.ndbr
2397 sudiksha 20  0 772152 40846 18700 5 1.3 2.0 0.07.87 gnome-terminal-
2484 sudiksha 20  0 225924 4144 3512 8 0.7 0.2 0.09.28 top
1981 sudiksha 20  0 526628 1416 8704 5 0.3 0.7 0.02.36 sshd-daemon
2647 sudiksha 20  0 608446 10420 9200 5 0.3 0.4 0.00.30 sshd-idonttty-so
2175 sudiksha 20  0 603876 17412 18640 5 0.3 0.7 0.00.34 sshd-usercard
1 root      0  0 117308 17216 11160 5 0.0 1.0 0.00.92 systemd
2 root      0  0 0 0 0 0 0 0 0.00.00 kthreadd
3 root      0  0 0 0 0 0 0 0 0.00.00 pool_workqueue_
4 root      0  20 0 0 0 0 0 0 0.00.00 kworker/R-rcu_
5 root      0  20 0 0 0 0 0 0 0.00.00 kworker/R-symc
6 root      0  20 0 0 0 0 0 0 0.00.00 kworker/R-slub
7 root      0  20 0 0 0 0 0 0 0.00.00 kworker/R-netns
11 root     0  20 0 0 0 0 0 0 0.00.00 kworker/0:0-events_highr
13 root     0  0 0 0 0 0 0 0 0.00.00 rcu_tasks_kthre
14 root     0  0 0 0 0 0 0 0 0.00.00 rcu_tasks_rude
15 root     0  0 0 0 0 0 0 0 0.00.00 rcu_tasks_trace
16 root     0  0 0 0 0 0 0 0 0.00.00 ksoftirqd/R
17 root     0  0 0 0 0 0 0 0 0.00.00 rcu_preempt
18 root     0  0 0 0 0 0 0 0 0.00.00 rcu_exp_gp_0
19 root     0  0 0 0 0 0 0 0 0.00.00 rcu_exp_gp_0_kthr
20 root     14  0 0 0 0 0 0 0 0.00.00 migration/0
```

```
[sudiksha@localhost Documents]$ free -t
              total        used        free      shared  buff/cache   available
Mem:           1748620        869960       395668        14068       653616       878660
Swap:           2097148           3096       2094052
Total:          3845768       873056       20489720
[sudiksha@localhost Documents]$ top
```

```
sudiksha@localhost:~/Documents
[sudiksha@localhost Documents]$ ps -e
PID TTY          TIME CMD
1 ?            00:00:00 systemd
2 ?            00:00:00 kthreadd
3 ?            00:00:00 pool_workqueue_
4 ?            00:00:00 kworker/R-rcu_
5 ?            00:00:00 kworker/R-symc
6 ?            00:00:00 kworker/R-slub
7 ?            00:00:00 kworker/R-netns
9 ?            00:00:00 kworker/0:0-events_highr
11 ?           00:00:00 kworker/R-mem_po
13 ?           00:00:00 rcu_tasks_kthre
14 ?           00:00:00 rcu_tasks_rude
15 ?           00:00:00 rcu_tasks_trace
16 ?           00:00:00 ksoftirqd/R
17 ?           00:00:00 rcu_preempt
18 ?           00:00:00 rcu_exp_gp_0
19 ?           00:00:00 rcu_exp_gp_0_kthr
20 ?           00:00:00 migration/0
21 ?           00:00:00 idle_inject/0
22 ?           00:00:00 cpuhp/0
23 ?           00:00:00 kconfta
26 ?           00:00:00 kworker/R-tnet
27 ?           00:00:00 kauditd
28 ?           00:00:00 khungtaskd
29 ?           00:00:00 oom_reaper
30 ?           00:00:00 kworker/R-write
31 ?           00:00:00 kcompact0
32 ?           00:00:00 ksm
33 ?           00:00:00 khungpaged
```

```
sudiksha@localhost:~/Documents
time ps -o 00:00:00 ps
[sudiksha@localhost Documents]$ vmstat
procs-----memory-----swap-----io-----system-----cpu-----
r b swpd free buff cache si so bi bo in cs us sy id wa st
1 0 3066 391320 2054 653124 0 2 1484 129 182 351 1 7 92 0 0

[sudiksha@localhost Documents]$ df
Filesystem      1K-blocks    Used Available Use% Mounted on
devtmpfs         840288      0    840288   0% /dev
tmpfs             874380      0    874380   0% /dev/shm
tmpfs            508774    5568    508216   1% /run
/dev/mapper/cs-root 17755160 6689740 11065412 38% /
/dev/sda1        98340 448748 59452 40% /boot
tmpfs            174860    100    174760   1% /run/user/1000
dev/sr0          52196 52196   0 100% /run/media/sudiksha/VBox_GAs_7.0.12

[sudiksha@localhost Documents]$ ping
ping: usage error: Destination address required
[sudiksha@localhost Documents]$ ping 172.16.4.1
PING 172.16.4.1 (172.16.4.1) 56(84) bytes of data:
^C
--- 172.16.4.1 ping statistics ---
126 packets transmitted, 0 received, 100% packet loss, time 128042ms

[sudiksha@localhost Documents]$ ifconfig
enp0s3: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.0.2.15 netmask 255.255.255.0 broadcast 10.0.2.255
    inet6 fe80::a00:27ff:fe4c:b833 prefixlen 64 scopeid 0x20<link>
    ether 08:00:27:4c:b8:33 txqueuelen 1000 (Ethernet)
    RX packets 163 bytes 15422 (15.0 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 34 bytes 3408 (3.3 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 34 bytes 3408 (3.3 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

```
sudiksha@localhost:~/Documents
/dev/sr0          52196 52196   0 100% /run/media/sudiksha/VBox_GAs_7.0.12
[sudiksha@localhost Documents]$ ping
ping: usage error: Destination address required
[sudiksha@localhost Documents]$ ping 172.16.4.1
PING 172.16.4.1 (172.16.4.1) 56(84) bytes of data.
^C
--- 172.16.4.1 ping statistics ---
126 packets transmitted, 0 received, 100% packet loss, time 128042ms

[sudiksha@localhost Documents]$ ifconfig
enp0s3: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.0.2.15 netmask 255.255.255.0 broadcast 10.0.2.255
    inet6 fe80::a00:27ff:fe4c:b833 prefixlen 64 scopeid 0x20<link>
    ether 08:00:27:4c:b8:33 txqueuelen 1000 (Ethernet)
    RX packets 163 bytes 15422 (15.0 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 34 bytes 3408 (3.3 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 34 bytes 3408 (3.3 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```